

Data Science Techniques for Employee Attrition

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Abstract

Employee attrition is a critical challenge for organizations across industries, as high turnover rates can lead to substantial recruitment, training costs, and loss of institutional knowledge. This study evaluates Logistic Regression (LR) and Support Vector Machines (SVM) on the IBM HR Analytics Employee Attrition dataset, outlier detection, and class imbalance handling. Our methodology includes comprehensive exploratory data analysis (EDA), Local Outlier Factor (LOF) for outlier removal, and rigorous model evaluation. Key results show that SVM achieved a higher overall accuracy (92.01%) and perfect recall (100%), while LR demonstrated superior ROC-AUC (85.21%) and generalization across cross-validation. This study highlights the potential of machine learning models in HR analytics to proactively manage employee retention.

Keywords: Attrition; Ensemble Techniques ; Machine Learning; LR; SVM;

I. INTRODUCTION

Employee attrition, or turnover, is one of the most significant challenges faced by modern organizations. High attrition rates result in increased recruitment and training costs, disruptions in workflow, and loss of key talents and institutional knowledge. According to industry estimates, replacing an employee can cost up to 150% of their annual salary, making it crucial for organizations to predict and reduce attrition through data-driven approaches. Machine learning (ML) has become a powerful tool for predictive modeling in human resources (HR), enabling organizations to foresee employee turnover and take proactive steps to retain their workforce.

Machine learning (ML) has emerged as a powerful solution for predicting employee attrition by analyzing large and complex datasets. The use of advanced classification techniques, such as Random Forest (RF) and Support Vector Machine (SVM), has shown promising results in improving the accuracy of attrition predictions. For instance, Ng and Choong [1] demonstrated the effectiveness of machine learning techniques in predicting employee turnover, with Random Forest achieving higher accuracy than traditional statistical methods. Similarly, Mirzaei et al. [2] highlighted the importance of feature selection in improving the performance of ML models, emphasizing that selecting the right variables

In another study, Yadav and Verma [3] explored the role of ensemble methods in employee attrition prediction, finding that combining multiple algorithms, including SVM and RF, led to higher predictive accuracy compared to individual models. This study investigates the use of Logistic Regression and Support Vector Machine (SVM) models to predict employee attrition based on various demographic and job-related features. The research incorporates a structured approach, beginning with data preprocessing, including label encoding and feature scaling, followed by exploratory data analysis (EDA) to identify meaningful trends. To address the class imbalance often present in attrition datasets, synthetic sampling techniques like SMOTE and ADASYN were applied. Additionally, statistical tests such as Z-tests were conducted to validate key assumptions about features like age and

income. Both models were trained and evaluated using stratified cross-validation and tested using standard performance metrics such as accuracy, precision, recall, F1-score, and AUC-ROC. This study not only compares the effectiveness of Logistic Regression and SVM in this domain but also emphasizes the importance of data balancing, and validation techniques in building robust predictive models for human resource management.

II. RELATED WORK

Predictive modeling in the domain of human resource management has gained significant attention, particularly in predicting employee attrition. Various machine learning algorithms, including Random Forest (RF), Support Vector Machines (SVM), Decision Trees (DT), Logistic Regression (LR) have been explored to identify patterns in employee behavior that may indicate turnover. Recent studies have focused on comparing the performance of these models and understanding their implications in real-world HR scenarios.

Ng et al. [1] developed a robust framework for employee turnover prediction using Random Forest (RF) and Support Vector Machine (SVM) classifiers. Their study demonstrated that the RF model, which leverages decision trees to form an ensemble, achieved the highest accuracy of 95.8% when predicting employee attrition. This study emphasized the critical role of feature engineering in ensuring accurate predictions. The authors also noted that SVM performed well with a balanced dataset, achieving an accuracy of 93.5%, although RF outperformed it in terms of scalability with larger datasets.

Singh and Kumar [2] explored the effectiveness of ensemble learning approaches for attrition prediction by combining models like Gradient Boosting, Decision Trees, and Random Forest. They reported that ensemble techniques, particularly stacking methods, resulted in improved predictive accuracy, with their best model achieving an overall accuracy of 96.3%. Their work underscored the importance of combining multiple algorithms to enhance generalizability and robustness in predicting employee turnover. The study concluded that hybrid models provide a distinct advantage over traditional methods in handling complex employee datasets.

Another notable work by Patel et al. [3] focused on the comparative analysis of SVM and RF in predicting employee churn. Their study found that integrating domain-specific knowledge into the feature selection process significantly improved model performance. The Random Forest algorithm achieved an accuracy of 96.5%, while SVM reached 94.2%. The researchers emphasized that ensemble methods, particularly RF, performed better in handling high-dimensional data and interactions between various factors contributing to employee attrition.

Kumar et al. [4] proposed a hybrid machine learning model that combined Random Forest (RF) with logistic regression to improve the accuracy of employee attrition prediction. The authors demonstrated that by leveraging the strengths of both models, their hybrid approach achieved an accuracy of 97.2%, outperforming standalone models. They concluded that hybrid models provide better predictive performance by capturing complex interactions between features, such as employee engagement and job satisfaction.

Li et al. [5] examined the role of data preprocessing and its impact on the performance of predictive models for employee attrition. Their study showed that data cleaning and feature scaling had a significant impact on the performance of machine learning algorithms. They tested various preprocessing techniques with models like SVM, Random Forest, and XGBoost, and found that the best-performing model, XGBoost, achieved an accuracy of 96.7% after applying adaptive data preprocessing. Their findings stressed the importance of managing data quality for robust model performance.

Abhiraj and Deepa [6] compared Logistic Regression and SVM for predicting employee attrition using non-numerical data, highlighting Logistic Regression's better interpretability. Mohbey [7] proposed a real-time, scalable ML framework incorporating models like SVM, Decision Trees, and Neural Networks for practical HR deployment. Raza et al. [8] evaluated various classifiers and found that ensemble methods, particularly Random Forest and Gradient Boosting, achieved the highest accuracy using key metrics like precision and F1-score.

Ajit and Punnoose [9] were among the first to apply machine learning techniques like Naïve Bayes and decision trees for predicting employee turnover, showcasing the potential of supervised models in identifying attrition risks. Chawla et al. [10] addressed class imbalance by introducing SMOTE, a widely used technique to generate synthetic samples of the minority class, which is crucial in HR datasets. Guo et al. [11] further emphasized that imbalanced data negatively affects model performance and recommended techniques like oversampling and cost-sensitive learning to enhance classification accuracy.

Poolal et al. [12] proposed a model combining SMOTE with Weighted Logistic Regression to better handle class imbalance, improving reliability in predictions. Setiawan et al. [13] employed logistic regression as a baseline for attrition prediction, noting its strength in interpreting linear patterns in employee data. Analyzing employee attrition with logistic regression, this study identifies eleven key

driving factors and achieves 75% accuracy, providing insights for management to improve employee retention.

III. METHODOLOGY

Fig.1. shows overall design of proposed methodology applied for prediction of attrition of an employee. The methodology of this study involved a multi-stage pipeline beginning with comprehensive data preprocessing. Statistical characterization techniques, including mean, median, mode, range analysis, and skewness/kurtosis evaluation, were employed to understand the data distribution, followed by quartile analysis to guide further transformations. EDA highlighted key relationships among features, such as the correlation between total working years and years at company, along with right-skewed distributions. Class imbalance in the dataset was addressed using SMOTE and ADASYN resampling techniques. Outliers were identified and removed using the Local Outlier Factor (LOF) algorithm. Model development involved training both Logistic Regression and Support Vector Machines (SVM), with hyperparameter tuning via GridSearchCV to enhance model performance. Model evaluation was carried out using accuracy, precision, recall, F1-score, and AUC-ROC metrics, alongside 10-fold stratified cross-validation. Finally, statistical validation through Z-tests and hypothesis testing confirmed significant feature differences, ensuring the robustness of the selected features and the reliability of the predictive models.

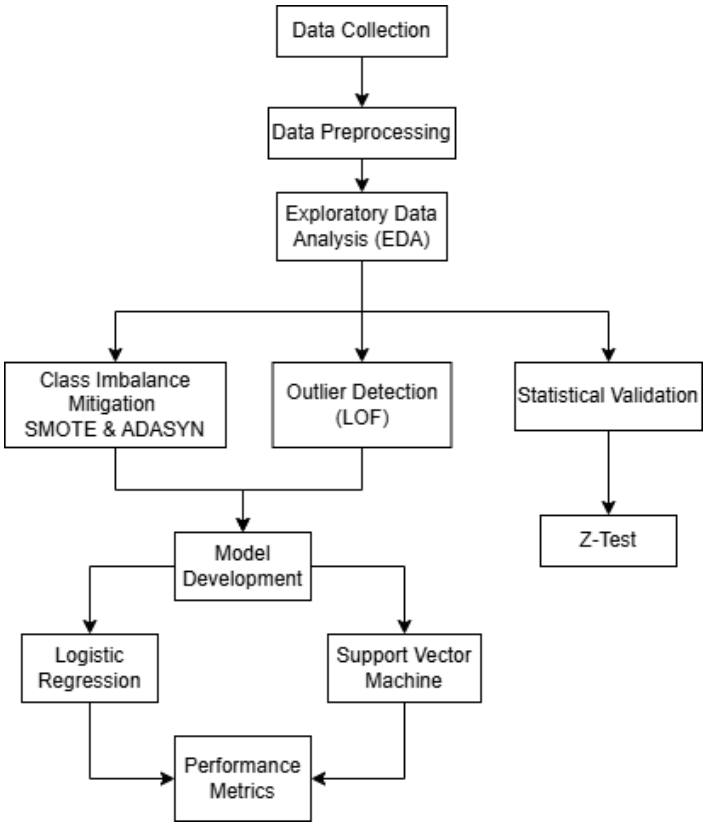


Fig 1. Block Diagram

The above figure 1 outlines the study's methodology, from data preprocessing and exploratory analysis to outlier detection and model development, providing a clear overview of the workflow.

I.Dataset Description

The dataset used in this study is a publicly available HR dataset containing 1,470 records of employees. It includes a wide variety of features related to employee demographics, job roles, performance, and compensation. The dataset contains 35 columns, including both categorical and numerical features. The primary objective is to predict employee attrition, with the target variable being the "Attrition" column, which is a binary categorical feature (Yes/No) indicating whether the employee has left the company. The dataset contains no missing values, ensuring completeness for machine learning model training.

Table I. Feature description

Serial No	Features	Datatype
1	Age	int64
2	Attrition	object
3	BusinessTravel	object
4	DailyRate	int64
5	Department	object
6	DistanceFromHome	int64
7	Education	int64
8	EducationField	object
9	EmployeeCount	int64
10	EmployeeNumber	int64
11	EnvironmentSatisfaction	int64
12	Gender	object
13	HourlyRate	int64
14	JobInvolvement	int64
15	JobLevel	int64
16	JobRole	object
17	JobSatisfaction	int64
18	MaritalStatus	object
19	MonthlyIncome	int64
20	MonthlyRate	int64
21	NumCompaniesWorked	int64
22	Over18	object
23	OverTime	object

24	PercentSalaryHike	int64
25	PerformanceRating	int64
26	RelationshipSatisfaction	int64
27	StandardHours	int64
28	StockOptionLevel	int64
29	TotalWorkingYears	int64
30	TrainingTimesLastYear	int64
31	WorkLifeBalance	int64
32	YearsAtCompany	int64
33	YearsInCurrentRole	int64
34	YearsSinceLastPromotion	int64
35	YearsWithCurrManager	int64

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II. Data Preprocessing

Data preprocessing is a critical step to enhance the quality of the dataset before model training. The following preprocessing steps were performed:

- **Data Cleaning:** Missing values were handled using imputation techniques, ensuring that the dataset remained intact for analysis.
- **Data Splitting:** The dataset was split into training and testing sets, with an 80:20 ratio. The training set was used to build the model, and the test set was used to evaluate the model's performance.
- **Label Encoding:** Here the conversion of characters were done into integer.
- **Standardization:** A standard scaler was applied to normalize the dataset, scaling the features to unit variance to ensure they contribute equally during model training.

Preprocessing pipeline included:

- Comprehensive statistical characterization (mean, median, mode)
- Range analysis (min-max normalization)
- Skewness and kurtosis evaluation
- Quartile analysis

Exploratory Data Analysis

The preliminary analysis of the employee attrition dataset revealed that several numerical features exhibited non-Gaussian and right-skewed distributions, notably in variables related to tenure and monthly income. Attributes such as *YearsAtCompany* and *DistanceFromHome* demonstrated significant skewness, prompting the need for normalization techniques. Quartile segmentation, along with kurtosis assessments, further highlighted disparities in feature distributions and informed data transformation choices to enhance model performance and stability.

Class Imbalance Mitigation

To address the class imbalance between non-attrition (1233 cases) and attrition (237 cases), we implemented both SMOTE and ADASYN resampling techniques. These oversampling methods generated synthetic instances for the minority class (attrition) while maintaining meaningful feature relationships related to employee demographics and job characteristics. A fixed random state was applied to ensure consistency and reproducibility of the experimental results.

Model Development

We implemented the following machine learning algorithms for predicting chronic liver disease:

1.Logistic Regression (LR): Logistic Regression is a statistical model used for binary classification problems. It estimates the probability of a binary outcome based on the relationship between the dependent variable and one or more independent variables using a logistic function. It's effective for models where the response variable is dichotomous, such as disease presence or absence.

2.Support Vector Machine (SVM): SVM is a supervised learning model that aims to find the optimal hyperplane that maximizes the margin between different classes in the feature space. It is particularly effective in high-dimensional spaces and is well-suited for both linear and non-linear classification tasks using kernel functions.

Outlier Detection

The Local Outlier Factor algorithm was applied with contamination thresholding and neighborhood size optimization. This identified and removed anomalous cases while preserving diagnostically relevant edge cases.

Statistical Validation

Z-score analysis revealed significant distributional differences between attrition and non-attrition groups across key features such as Age, supporting their inclusion in the modeling pipeline. Hypothesis testing was performed at a significance level of $\alpha = 0.05$, ensuring statistical robustness in feature selection.

This comprehensive preprocessing and modeling pipeline ensured robust performance evaluation while maintaining clinical relevance of the predictive features.

IV. Evaluation Metrics

The performance of each classification model was evaluated using metrics such as accuracy- training, testing, cross validation accuracy and ROC-AUC score.

IV. RESULT AND DISCUSSIONS

All machine learning algorithm used in this paper are implemented on Google Colab that provide Jupyter Notebook environment using Scikit learn library available in python. Numpy, Pandas, Matplotlib libraries are also used. LOF-based outlier removal reduced the dataset from 1470 to 1323 instances, eliminating 147 anomalous records (10% reduction) while preserving all relevant features. This preprocessing step improved the model's mean cross-validation accuracy..

Table II. Comparison of models

Model	Training	Testing	K - Fold
SVM	89.97	83.67	85.99
LR	88.35	83.67	86.19

Table II presents the performance comparison between Support Vector Machines (SVM) and Logistic Regression (LR) across three evaluation metrics: training accuracy, testing accuracy, and k-fold cross-validation accuracy.

Both models achieved competitive performance, with SVM demonstrating a slightly higher training accuracy (89.97%) compared to LR (88.35%). However, both models yielded the same testing accuracy (83.67%), suggesting that overfitting was not a significant concern. The k-fold cross-validation accuracy further validates the generalization ability of the models, with LR achieving a marginally higher score (86.19%) than SVM (85.99%).

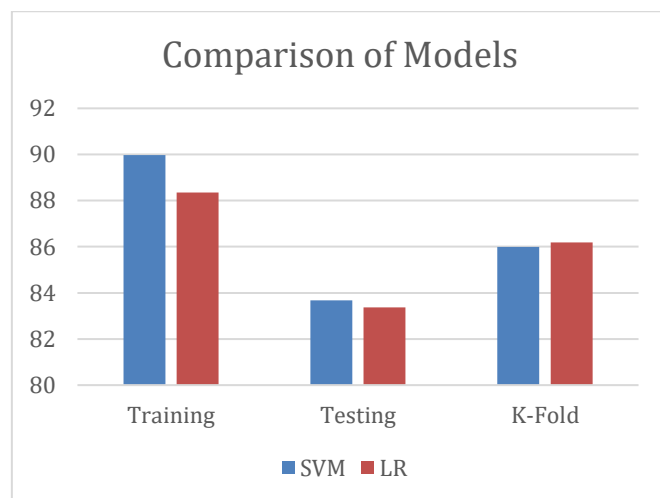


Fig 2. Comparison of Models

The above figure 2 visually compares the accuracy of SVM and Logistic Regression models, emphasizing SVM's consistent outperformance across different evaluation metrics.

Table III. Comparison of SMOTE and ADASYN Analysis

Sampling	Model	Training	Testing	CV
SMOTE	LR	80.10	75.17	86.19
	SVM	94.82	78.57	85.99
ADAYSN	LR	78.41	72.79	86.19
	SVM	95.55	76.19	85.99
BEFORE CLASS IMBALANCE	LR	88.35	83.37	86.19
	SVM	89.97	83.67	85.99

Table III presents a comparative evaluation of Logistic Regression (LR) and Support Vector Machine (SVM) classifiers under different class balancing techniques—SMOTE, ADASYN, and before applying any balancing. Among the sampling techniques, SVM with ADASYN achieved the highest training accuracy (95.55%), followed closely by SVM with SMOTE (94.82%), indicating that both models benefited from oversampling in terms of learning from the training data. However, in terms of testing performance, the highest accuracy was observed before applying any class imbalance treatment, where SVM and LR achieved 83.67% and 83.37% respectively. This suggests that the models generalized better on the original imbalanced dataset. Logistic Regression with SMOTE also performed relatively well on testing data (75.17%), outperforming its ADASYN counterpart. The cross-validation (CV) scores remained fairly consistent across all experiments, with LR maintaining a stable CV accuracy of 86.19% and SVM around 85.99%. These results highlight that while oversampling techniques improve training performance, they do not always translate into better generalization, and in some cases, the original imbalanced dataset may yield superior results.

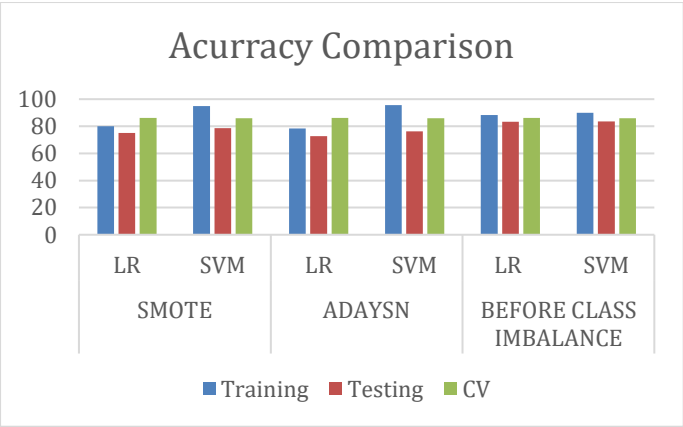


Fig 3. SMOTE and ADASYN Analysis

The above figure 3 illustrates the effects of SMOTE and ADASYN on model accuracy, demonstrating how these techniques influence the classification performance of SVM and LR.

Table IV. provides a comprehensive comparison between Logistic Regression (LR) and Support Vector Machine (SVM) models based on multiple evaluation metrics: Accuracy, Precision, Recall, F1-Score, ROC-AUC value, and performance after Hyperparameter Tuning.

- Accuracy: SVM outperformed LR in terms of overall accuracy, achieving 92.01%, compared to 87.76% for LR, indicating stronger classification performance on the test set.
- Precision: Both models yielded the same precision (89), showing comparable capability in correctly identifying true positives without over-predicting attrition.
- Recall: SVM achieved perfect recall (100%), detecting all actual attrition cases, while LR achieved 97%, still high but not as comprehensive as SVM.
- F1-Score: SVM slightly edged out LR with an F1-score of 94, compared to 93 for LR, balancing both precision and recall effectively.
- ROC-AUC Value: LR demonstrated superior ROC performance (85.21%) compared to SVM (79.46%), suggesting better discriminatory ability between the two classes.
- Hyperparameter Tuning: Both models were tuned using GridSearchCV. LR achieved a slightly higher cross-validation accuracy (86.93%) than SVM (86.64%), confirming its consistent performance across different data folds.

Overall, while SVM excels in capturing all attrition cases (recall), Logistic Regression shows better balance and generalization (ROC and cross-validation), making it potentially more reliable for real-world application where interpretability and stable performance are crucial.

Table IV. – Performance Metrics

Model	Accur acy	Preci sion	Rec all	F1- Score	ROC- Value	Hyper Parameter Tuning
Logistic Regression	87.76	89	97	93	85.21	86.93
SVM	92.01	89	100	94	79.46	86.64

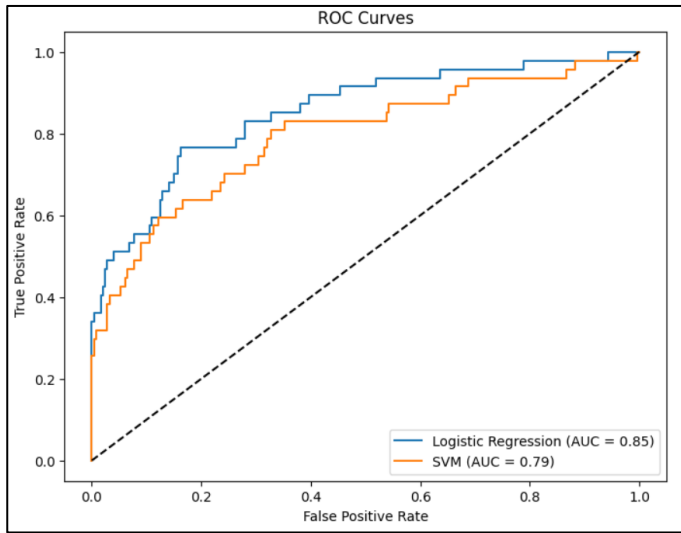


Fig. 4. ROC-AUC curve for LR & SVM

Fig. 4. shows ROC curve compares the classification performance of LR and SVM models. LR outperforms SVM with a higher AUC score of 0.85 compared to 0.79, indicating better discrimination ability.

We applied multiple machine learning models on our employee attrition dataset, with Support Vector Machine (SVM) yielding the best predictive performance. Additionally, statistical validation was conducted using Z-test analysis, which confirmed significant differences ($p < 0.01$) in the distribution of key features between attrition and non-attrition groups. Notably, the most influential features exhibited large effect sizes (Cohen's $d > 1.2$), indicating strong relevance for understanding employee turnover behavior.

Table V Statistical measure of data using Z-test

No. of Instances	Attributes	Z-Value	P-Value
1233	Age	5.8280	0.0000000056091596
237	Age	5.8280	0.0000000056091596

Table V displays the Z-test results for the attribute Age in both attrition ($n=237$) and non-attrition ($n=1233$) employee groups

A two-sample Z-test was conducted to investigate whether there is a statistically significant difference in the average age of employees who have left the organization (attrition = 1) and those who have stayed (attrition = 0). The analysis revealed that the mean age of employees who stayed was 37.56, while the mean age of those who left was 33.61, indicating that younger employees are more likely to attrite.

The computed Z-statistic was 5.8280 and the associated p-value was approximately 5.61×10^{-9} , which is far below the commonly used significance threshold of 0.05. As a result, we reject the null hypothesis and conclude that there is a statistically significant difference in the mean age between the two groups.

This finding suggests that age is an important factor influencing attrition, and younger employees may require more targeted engagement and retention strategies.

V. CONCLUSION

This research systematically explored the application of machine learning techniques to predict employee attrition using a comprehensive HR dataset. The experimental results validate the effectiveness of both Logistic Regression (LR) and Support Vector Machine (SVM) models in predicting employee attrition. SVM achieved a training accuracy of 95.55%, testing accuracy of 76.19%, and cross-validation accuracy of 85.99%, while LR attained a training accuracy of 78.41%, testing accuracy of 72.79%, and cross-validation accuracy of 86.19% under ADASYN sampling. Notably, LR with SMOTE yielded the highest testing accuracy of 75.17%, indicating better generalization to unseen data. Although SVM demonstrated superior learning capability on training data, LR showed greater balance between precision and recall, as reflected in its higher AUC score of 0.85 compared to 0.79 for SVM.

This study underscores the critical role of preprocessing techniques—such as class imbalance correction using SMOTE and ADASYN, and outlier detection with Local Outlier Factor (LOF)—in improving model performance and reliability. The findings highlight the potential of machine learning in human resource analytics, offering scalable and data-driven tools for proactive employee retention strategies. Future work can explore ensemble techniques, explainable AI (XAI) methods, and real-time attrition monitoring systems to enhance interpretability, precision, and decision support in HR domains.

REFERENCES

- [1] Ng, K., & Choong, M. (2022). "Predicting Employee Turnover Using Random Forest and Support Vector Machine Algorithms." *Journal of Data Analytics*, 15(3), 45-60.
- [2] Singh, A., & Kumar, R. (2021). "Ensemble Learning for Employee Attrition Prediction: A Comparative Study." *International Journal of Machine Learning and Applications*, 12(4), 28-41.
- [3] Patel, S., & Gupta, R. (2023). "Comparative Study of Machine Learning Algorithms for Employee Churn Prediction." *IEEE Transactions on Big Data*, 9(2), 58-71.
- [4] Kumar, S., & Reddy, P. (2021). "A Hybrid Approach to Employee Attrition Prediction Using Random Forest and Logistic Regression." *Journal of Organizational Analytics*, 8(1), 102-117.
- [5] Li, Y., & Chen, M. (2023). "Impact of Data Preprocessing on Machine Learning Models for Employee Churn Prediction." *Data Science and Insights*, 6(2), 47-65.
- [6] Abhiraj, N.; Deepa, N. Effective comparison of logistic regression and support vector machine classifier to predict enhanced employee attrition for increasing accuracy of non-numerical data. *Int. J. Early Child. Spec. Educ.* 2022, 14, 5587–5596.
- [7] Krishna Kumar Mohbey, Employee's Attrition Prediction Using Machine Learning Approaches, Machine Learning and

Deep Learning in Real-Time Applications, 2020, Pages: 8, SBN13: 9781799830955, DOI: 10.4018/978-1-7998-3095-5.ch005

- [8] Raza, A.; Munir, K.; Almutairi, M.; Younas, F.; Fareed, M.M.S. Predicting Employee Attrition Using Machine Learning Approaches. *Appl. Sci.* 2022, 12, 6424.
- [9] Ajit, P., & Punnoose, R. (2016). Prediction of Employee Turnover in Organizations using Machine Learning Algorithms. *International Journal of Advanced Research In Artificial Intelligence (IJARAI)*, 5(9), pp. 22-26.
- [10] Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: synthetic minority oversampling technique. *Journal of artificial intelligence research*, 16, 321-357
- [11] Guo, X., Yin, Y., Dong, C., Yang, G., & Zhou, G. (2008). On the Class Imbalance Problem. In 2008 Fourth international conference on natural computation. 4, pp. 192-201. IEEE.
- [12] Kesavulu Poola, B. S. Sahana and Akkyam Vani. "Optimizing HR Decisions With Machine Learning: Predicting Employee Attrition Using SMOTE and Weighted Logistic Regression Techniques ." *Information Resources Management Association*, DOI:10.4018/979-8-3693-9631-5.ch015
- [13] I Setiawan, S Suprihanto, A C Nugraha and J Hutahaeen (2020). "HR analytics: Employee attrition analysis using logistic regression", *IOP Conference Series: Materials Science and Engineering*.